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Report On

“Humanized Vision – Real-Time Emotion Detector & SOS Alert System”

Bachelor Of Engineering in Artificial Intelligence & Machine Learning

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ABSTRACT

In today's rapidly evolving technological environment, artificial intelligence is increasingly being used to improve human safety and wellbeing. One such application is the detection of human emotions through facial expressions. Emotional recognition systems can play a crucial role in identifying distress or dangerous situations where a person may require immediate assistance.

The Humanized Vision – Real-Time Emotion Detector & SOS Alert System is designed to monitor a user's facial expressions using a webcam and identify emotional states in real time. The system uses machine learning models to detect emotions such as happiness, sadness, anger, surprise, fear, neutral, and disgust.

The primary goal of this project is to enhance personal safety by detecting fearful emotional states continuously. When the system detects fear for a sustained duration, it automatically triggers an SOS alert mechanism. This alert includes the user's location and emergency contact details, enabling faster assistance during potential emergency situations.

The system uses face-api.js, a JavaScript-based deep learning library that performs facial detection, landmark detection, and emotion classification directly in the browser. Since all computations are executed locally, the system ensures user privacy by avoiding storage or transmission of facial data.

This project demonstrates how artificial intelligence, computer vision, and web technologies can be integrated to create intelligent safety monitoring systems that can assist individuals in critical situations.

INTRODUCTION

In modern society, personal safety and rapid emergency response have become major concerns. Many individuals may face situations where they are unable to manually call for help due to stress, fear, or physical limitations. Artificial Intelligence provides an opportunity to develop intelligent systems that can recognize distress and automatically trigger emergency alerts.

Facial expressions are one of the most reliable indicators of human emotions. By analyzing facial features using computer vision techniques, machines can identify emotional states such as happiness, sadness, anger, fear, and surprise. These technologies are widely used in applications such as human–computer interaction, mental health monitoring, and security systems.

The **Humanized Vision system** uses a webcam to monitor facial expressions in real time and detect emotional states using deep learning models. The system continuously analyzes the user's facial expressions and determines the dominant emotion with a confidence score.

If the system detects **fearful emotion continuously for a certain period**, it assumes that the user may be in danger and automatically activates an SOS alert. The alert system provides a countdown allowing the user to cancel the alert if it was triggered accidentally. Otherwise, the system sends an alert containing location information to predefined emergency contacts.

The system is implemented as a **web application using React, TypeScript, and face-api.js**, making it lightweight and accessible through modern browsers without requiring installation. This approach ensures both convenience and privacy since all processing occurs locally on the user's device.

DESCRIPTION ABOUT PROJECT

The **Humanized Vision – Real-Time Emotion Detector & SOS Alert System** is an intelligent web application designed to detect human emotions and provide emergency assistance when distress is identified.

The system operates using the following workflow:

1. The user starts the webcam through the application interface.
2. The system captures live video frames from the camera.
3. A face detection algorithm identifies the face in the frame.
4. Facial landmarks such as eyes, nose, and mouth are extracted.
5. The emotion recognition model analyzes facial expressions.
6. The system determines the dominant emotion and its confidence score.
7. If the emotion **fearful** is detected continuously with high confidence, an SOS alert is triggered.
8. The system retrieves the user's location using the browser's geolocation API.
9. An emergency message containing the detected emotion, time, and location is generated.
10. The alert is sent to emergency contacts stored in the system.

The application detects the following emotions:

- Happy
- Sad
- Angry
- Fearful
- Surprised
- Neutral
- Disgusted

The emotion detection is performed using pretrained deep learning models available in the **face-api.js library**. These models include:

- Tiny Face Detector
- Face Landmark Detection Model
- Face Expression Recognition Model

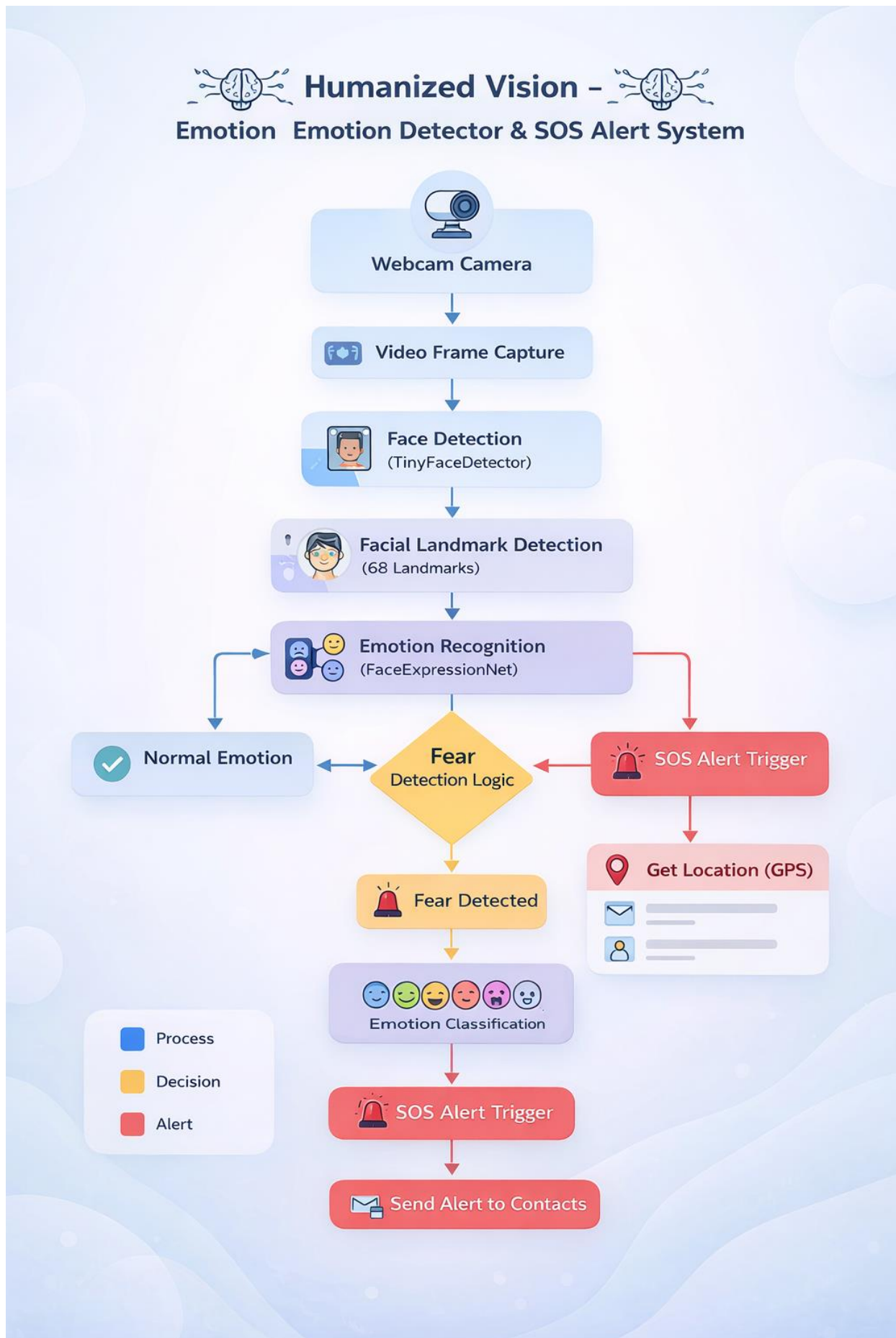
The system performs emotion detection approximately every **500 milliseconds**, ensuring real-time monitoring.

To improve usability and safety, the application also provides:

- Manual SOS trigger option
- Emergency contact management
- Event logging system
- Privacy protection by local processing

This project demonstrates how computer vision and artificial intelligence can be integrated into web-based systems to create real-time safety monitoring applications.

SYSTEM ARCHITECTURE / FLOW DIAGRAM



SOURCE CODE

```
import { useState, useRef, useEffect } from "react";
import { WebcamFeed } from "@components/WebcamFeed";
import { EmotionDisplay } from "@components/EmotionDisplay";
import { SOSAlert } from "@components/SOSAlert";

const Index = () => {

  const [currentEmotion, setCurrentEmotion] = useState({ emotion: "neutral", confidence: 0
});
  const [showSOSAlert, setShowSOSAlert] = useState(false);

  const fearStartTime = useRef(null);

  const handleEmotionDetected = (data) => {
    setCurrentEmotion(data);
  };

  const handleFearDetected = (confidence) => {
    if (confidence >= 0.6) {
      if (fearStartTime.current === null) {
        fearStartTime.current = Date.now();
      } else {
        const duration = Date.now() - fearStartTime.current;

        if (duration >= 2000) {
          setShowSOSAlert(true);
        }
      }
    } else {
      fearStartTime.current = null;
    }
  };

  return (
    <div>
      <h1>Humanized Vision – Emotion Detector</h1>

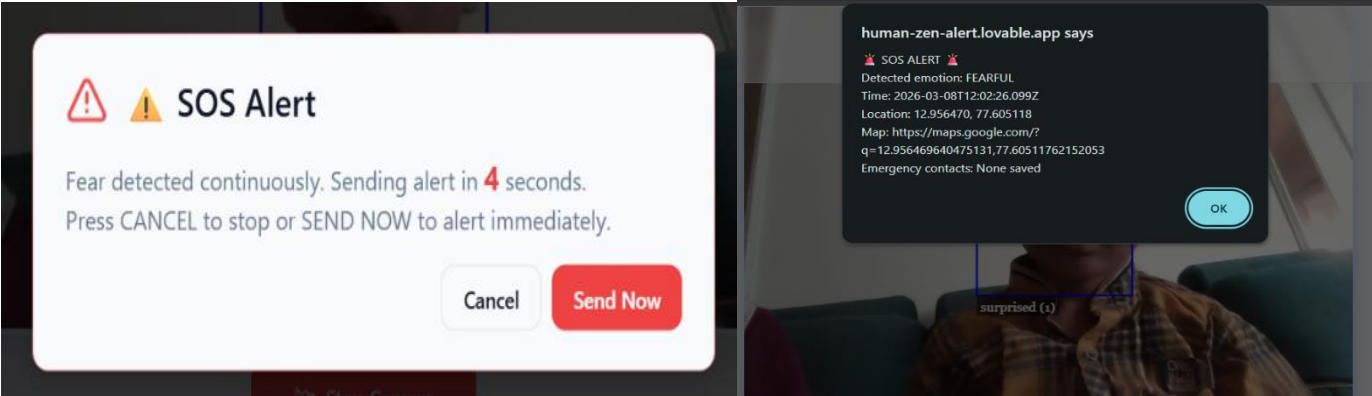
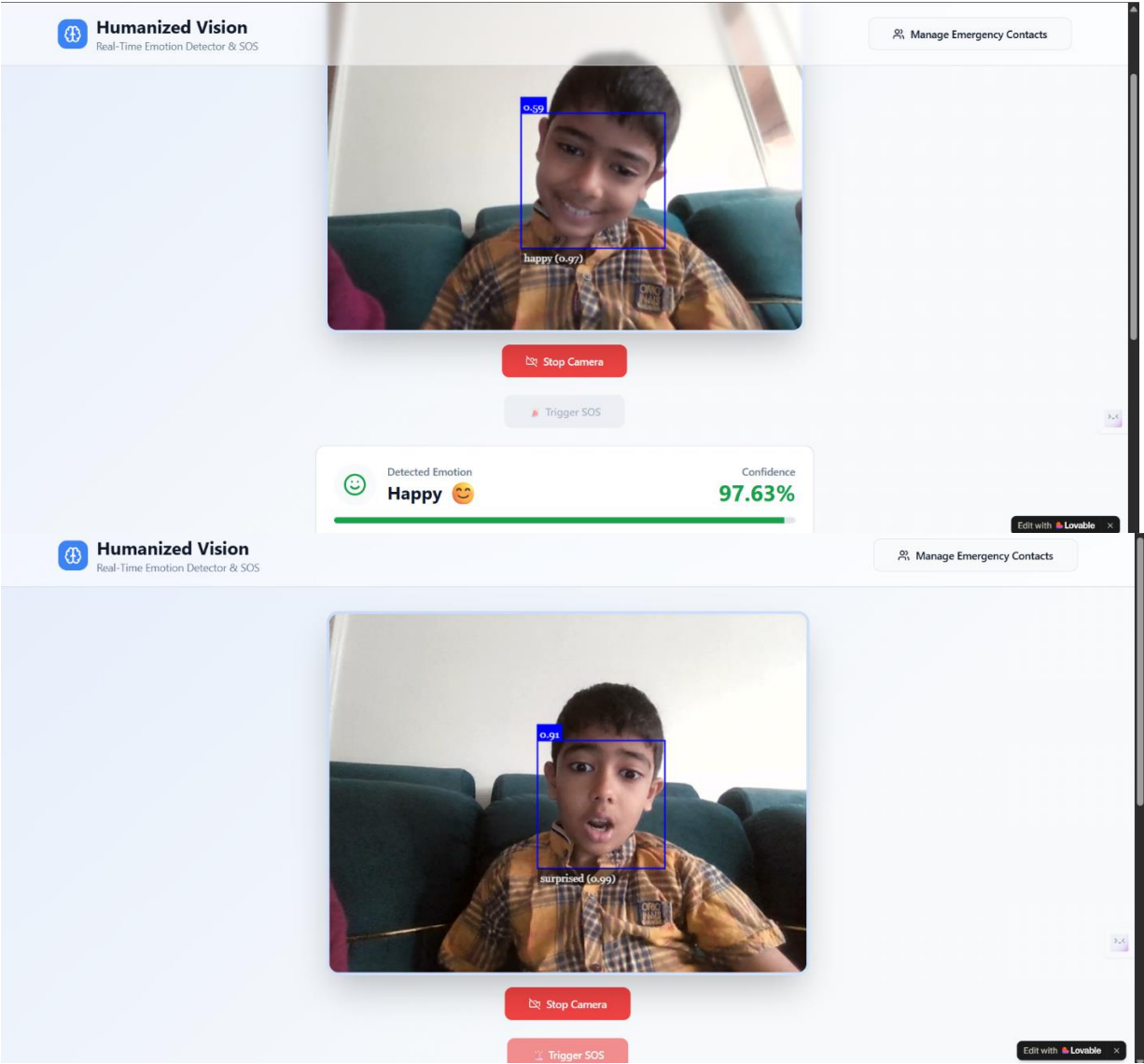
      <WebcamFeed
        onEmotionDetected={handleEmotionDetected}
        onFearDetected={handleFearDetected}
      />

      <EmotionDisplay
        emotion={currentEmotion.emotion}
        confidence={currentEmotion.confidence}
      />

      <SOSAlert isOpen={showSOSAlert} />
    </div>
  );
};

export default Index;
```

SCREENSHOT



CONCLUSION

The **Humanized Vision – Real-Time Emotion Detector & SOS Alert System**

successfully demonstrates the integration of artificial intelligence, computer vision, and web technologies to create a real-time emotion monitoring system.

The system detects human emotions using facial expression analysis and provides an automated emergency alert mechanism when distress is detected. By monitoring emotions such as fear, the system can identify potentially dangerous situations and assist users by triggering SOS alerts.

The use of **face-api.js and deep learning models** enables accurate emotion recognition directly within the browser. Since all processing occurs locally, the system ensures user privacy and prevents unnecessary storage or transmission of facial data.

This project highlights the potential of AI-based systems in improving **personal safety and emergency response**. It also demonstrates how modern web technologies such as React and TypeScript can be used to develop intelligent applications that are accessible, efficient, and scalable.

Future improvements may include integrating mobile notifications, SMS alerts, and advanced behavioral analysis to enhance the system's reliability and real-world applicability.